

Anaphora in Static Orthodox First-Order Logic

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Abstract. The paper demonstrates representing pronominal anaphora in the orthodox first-order predicate logic with the completely standard syntax and semantics. Specifically, we show how to represent not yet resolved pronouns in conventional logical formulas, and then record the result of their resolution (coindexing). Our proposal is static, simply-typed, minimalist, with the orthodox quantification, binding and scope. It relies on the 1920s logic, and applies classically and intuitionistically – in particular, treating implication as is. It is inspired by and extensively relies on intuitionistic Skolemization.

We demonstrate our approach on intra- and inter-sentential pronominal anaphora; bound-variable anaphora; conditional, relative-clause and proportional quantifier donkey sentences; quantificational subordination.

The approach aims to separate out pragmatic concerns from meaning composition: being able to map a syntactic structure to a logical formula representing the meaning, phrase-by-phrase and clause-by-clause, without prior coindexing and regardless of context. Pronoun resolution is to be done in the next, pragmatic step, based on the obtained formulas. Although coindexing is delegated to pragmatics, the logical formulas constrain the coindexing, helping the resolution and delivering predictions, of unavailability of particular antecedents.

1 Introduction

The paper is motivated by two problems, methodological and logical. Many existing semantic analysis frameworks either rely on prior full coindexing of anaphoric elements and their antecedents (see [5] for an earlier example), or perform the pronoun resolution in the process of building the logical formula representing truth conditions (see [1] for a more recent example), as was observed in [11]. Although coindexing may seem doable from the surface syntax, it is not always that easy and, generally, cannot be done without the context or world knowledge (see §7 for further discussion). Muskens put it well: “anaphora resolution is subject to semantic constraints,

that explaining these constraints is a crucial part of the theory, and that assuming inputs to come with correct indices therefore begs an important question.” [11].

We believe it is worth separating semantics from pragmatics – just as it is worth separating syntax from semantics, or segmentation from the parsing proper. The mapping from a syntactic structure to a logical formula representing its meaning can be done compositionally – if we free ourselves from concerns about pronoun resolution. The resulting logic formula may be a good structure on which to perform the pronoun resolution. Mainly, as the paper demonstrates, the obtained logic formula helps pronoun resolution by placing constraints on possible antecedents. We do obtain predictions, of unavailability of particular antecedents and particular readings.

We hence envision semantic analysis as a multi-stage process: first, deriving logical formulas, phrase-by-phrase, clause-by-clause. The formulas feed to the next phase: pronoun resolution (coindexing). Finally, the formulas are amended with the resolution result. This multi-stage process is to be contrasted with the ‘resolution-on-the-fly’ of DRT [8] and Muskens [11].

We are not proposing any new syntax-semantic interface or pronoun resolution. In fact, our proposal is compatible with many existing ones (e.g., QR, categorial grammar formalism, CCG, ACG, etc.). What we propose is how to represent not yet resolved pronouns in those formalisms, and how to later record the result of their resolution. In the simplest case, pronouns look as if they were proper names. However, these “proper names” depend on the universally quantified variables in scope. We also show how the formulas constrain the resolution and hence deliver antecedent unavailability predictions.

The motivation comes from the work of Butler [2], whose full-scale, scalable semantic framework also uses multi-stage process with a particular (graph-based) intermediate representation. Our aim is to understand his work in clear and strictly logical terms.

The logical problem is the so-called existential disclosure, clearly enunciated by Dekker [4, 5]: to explain anaphoric phenomena, an existential quantifier apparently needs to, in effect, bind variables beyond its lexical scope. Much work on dynamic logic (as well con-

tinuation, monad, etc. semantics) can be seen as developing a mechanism for the disclosure.

In this paper we observe that the existential disclosure can be achieved by Skolemization, which is the tried-and-true way to replace an existential with its witness. It was proposed back in 1920s and is widely used in logic meta-theory (e.g., in Henkin method for proving completeness) as well as in automated theorem provers.

All in all, we present a theory of:

- intra- and inter-sentential pronominal anaphora;
- bound-variable anaphora;
- conditional, relative-clause and proportional quantifier donkey sentences
- quantificational subordination

that is based *solely* on the orthodox predicate logic with the standard model-theoretic semantics. It is static, with the conventional notions of scope and binding. We explain the mechanism of anaphoric references in logical terms; give the orthodox intuitionistic account of discourse anaphora and donkey sentences; reproduce the empirical data (e.g., the absence of binding out of negated and universally quantified clauses).

We particularly stress an intuitionistically valid Skolemization, in which case a Skolem constant/function may truly be regarded as a witness, and hence properly serve as an antecedent. The intuitionistically valid Skolemization is also important to prevent overgeneration; see the particularly clear example of “not every” in §4.

We mostly restrict our attention to singular pronouns whose referent is a bound variable.

2 Warm-up: FOL, Satisfaction, Skolemization

This section introduces the notation and reminds the often used satisfaction and Skolemization. The logic is the standard classical first-order predicate logic (FOL); see [15] for reference. Terms are made from variables such as x , m and function symbols – such as **john** (of arity 0, also called constants) and **fatherOf** (of arity 1); note the font and the case. An atomic formula such as **Man(john)** or

$\text{Walked}(\text{fatherOf}(x))$ is an application of a predicate symbol (such as Man – note the case) to zero or more terms. Special, familiar syntax is used for atomic formulas representing term equality, such as $\text{john}=x$. General formulas (often notated ϕ and ψ) are built from atomic using the standard logical connectives and quantification: for example: $\exists x. \text{Man}(x) \wedge \text{Walk}(x)$. For brevity, we may write $\exists x. \text{P}(x) \wedge \phi$ as $\exists x \in \text{P}. \phi$ and $\forall x. \text{P}(x) \rightarrow \phi$ as $\forall x \in \text{P}. \phi$.

An *interpretation* $\mathcal{M} = (\mathcal{D}, \mathcal{I})$ of a logical formula ϕ is specified by a non-empty set $\mathcal{D}$ (domain of discourse) and an interpretation function $\mathcal{I}$ that maps every constant to a member of $\mathcal{D}$, every arity- n function symbol to a map $\mathcal{D}^n \mapsto \mathcal{D}$ and every arity- n predicate letter to a n -place relation over $\mathcal{D}$. A *variable assignment* $\mathcal{V}$ is a mapping of variables to elements of $\mathcal{D}$. *Satisfaction* is a relation between ϕ , $\mathcal{M}$, and $\mathcal{V}$ – notated as $\mathcal{M} \models_{\mathcal{V}} \phi$ and read as $\mathcal{M}$ satisfies ϕ (under the assignment $\mathcal{V}$, which is often left implicit).

As an example, $\mathcal{M} \models_{\mathcal{V}} \text{john}=x$ whenever $\mathcal{I}(\text{john}) = \mathcal{V}(x)$. In other words, whenever john and x are interpreted as the same domain element. Likewise, $\mathcal{M} \models_{\mathcal{V}} \text{Walk}(\text{john})$ whenever $\mathcal{I}(\text{john}) \in \mathcal{I}(\text{Walk})$. $\mathcal{M}$ satisfies an existential formula $\mathcal{M} \models_{\mathcal{V}} \exists x. \phi$ if and only if $\mathcal{M} \models_{\mathcal{V}'}$ ϕ for some assignment $\mathcal{V}'$ that agrees with $\mathcal{V}$ except possibly at the variable x . On the other hand, $\mathcal{M}$ satisfies $\mathcal{M} \models_{\mathcal{V}} \forall x. \phi$ just in case $\mathcal{M} \models_{\mathcal{V}'}$ ϕ for every $\mathcal{V}'$ that agrees with $\mathcal{V}$ except possibly at x .

Skolemization is the elimination of an existential: replacing a subformula $\exists x. \phi$ with ϕ in which every free occurrence of x is substituted with $\text{c}(y_1, \dots, y_k)$: so-called witness, or a Skolem term. Here, c is a fresh function symbol and $y_1, \dots, y_k$, often abbreviated as $\bar{y}_k$, $k \geq 0$, are all the variables in scope [16]. (This is a so-called Outer Skolemization [12]. In Inner Skolemization, $\bar{y}_k$ are the free variables in ϕ .)

For example, Skolemizing $\exists x. \text{Man}(x) \wedge \text{Walk}(x)$ gives $\text{Man}(\text{c}) \wedge \text{Walk}(\text{c})$. We write $\mathbb{S}[\psi]$ for a formula in which an existential subformula has been Skolemized; we call $\mathbb{S}[\psi]$ itself a Skolemization of ψ . The Skolemization theorem states that the formulas ψ and $\mathbb{S}[\psi]$ are *equi-satisfiable*: if there is an interpretation for one, then there is for the other.

An existentially-quantified subformula to Skolemize must not be under negation or at the left-hand-side of an implication. This also ensures that equi-satisfiability holds intuitionistically, and the

Skolem constant/function may truly be interpreted as a witness.¹ We deal with negation and implication in §5.

3 Pronoun Terms

The output of a syntax-semantic interface is a predicate logic formula expressing truth conditions. We now describe how to represent yet to be resolved pronouns in these formulas.

Definition 1. *A pronoun is represented by the term $f(x_1, \dots, x_n)$ ($n \geq 0$) where f is a fresh function symbol (constant) and $\bar{x}_n$ are the variables in scope.² We call such terms pronoun terms.*

For example,

- (1) A man walked. John saw him.
- (2) $(\exists x. \text{Man}(x) \wedge \text{Walk}(x)) \wedge \text{Saw}(f_1, \text{john})$

The meaning of the two-sentence discourse (1) is straightforwardly expressed by (2). Since the second sentence has no quantifiers or free variables, ‘him’ is represented by the constant: f_1 .

From the definition of the satisfiability, if an interpretation $(\mathcal{D}, \mathcal{I})$ satisfies (2) the following must be true:

$$\mathcal{I}(\text{Man})(d_0) \wedge \mathcal{I}(\text{Walk})(d_0) \wedge \mathcal{I}(\text{Saw})(d_1, \mathcal{I}(\text{john}))$$

where $d_0 \in \mathcal{D}$ is the satisfying interpretation of x for $\text{Man}(x) \wedge \text{Walk}(x)$ and $d_1 = \mathcal{I}(f_1)$. These d_0 and d_1 may or may not be the same. In other words, the satisfying interpretations of (2) may be partitioned into those that interpret f_1 and the bound variable x identically, and those that do not. The two choices corresponds to the range of meanings of (1): the antecedent of ‘him’ may be ‘a man’ – or the pronoun may remain unresolved. In the latter case, the pronoun may refer to something else in the context or common

¹ Incidentally, in (classical) automated theorem proving even stronger conditions are imposed: a formula to Skolemize must be in a prenex normal form (i.e., all quantifiers at the front), or in a negation normal form (i.e., containing only universal and existential quantification, disjunction and conjunction, with negation applied only to atomic formulas) [12].

² cf. Outer Skolemization [12].

knowledge, or just left to the speaker imagination: pragmatic use of pronouns, or exophora [13].

The Skolemized version of (2) is

$$(3) \quad (\text{Man}(c_1) \wedge \text{Walk}(c_1)) \wedge \text{Saw}(f_1, \text{john})$$

By equi-satisfiability, it has the same variety of interpretations as (2); in some of them the constants c_1 and f_1 are interpreted the same. Skolemization lets us select such interpretation through the formula itself (syntactically, so to speak), by adjoining closed equality formulas, to be called resolution restrictions:

$$(4) \quad (\text{Man}(c_1) \wedge \text{Walk}(c_1)) \wedge \text{Saw}(f_1, \text{john}) \wedge c_1 = f_1$$

Now, only the interpretations with ‘him’ referring to the walking man satisfy (4). The restricted formula is, of course, not equivalent to (3); rather, (4) implies (3). The resolution restriction $c_1=f_1$ is quite like the coindexing notation commonly used in semantic analyses.

A resolution restriction hence is the logical formula that represents the result of a pronoun resolution: it equates the pronoun term with some other, non-pronoun term. Since it is adjoined to the main logical form, it has to be closed. Concretely, a resolution restriction may take several forms:

- $\forall \bar{x}_n. f(\bar{x}_n) = x_l$ where x_l is one of $\bar{x}_n$. A satisfying interpretation then maps f to a projection function (when $n=1$, the identity function). As we shall soon see, this resolution restriction expresses bound-variable anaphora.
- $\forall \bar{x}_n. f(\bar{x}_n) = t$ where t is a ground term. The meaning of such f is the constant (function).
- $\forall \bar{x}_m. f(\bar{t}_n) = b(\bar{t}'_k)$ where $\bar{x}_m$ are free variables in $\bar{t}_n$, and all free variables in $\bar{t}'_k$ are included in $\bar{x}_m$. A particular case is $\forall \bar{x}_n. f(\bar{x}_n) = b(\bar{x}_n)$, meaning f is interpreted as the same function (constant, for $k=0$) as b : f co-refers with b .

The form a resolution restriction may *not* take is $\forall \bar{x}_m. f(\bar{t}_n) = t'$ where t' has a free variable not occurring in the pronoun term. Such a restriction is not closed and hence deemed ill-formed. Semantically, a resolution restriction is to constrain the interpretation of pronoun terms only, not of variables.

How exactly to resolve a pronoun – that is, how to come about the resolution restriction – is in the domain of pragmatics and out of scope for us. (It is in scope for Butler [2], who describes the procedure in more detail. See also §7 for further discussion.) In this paper we are interested in the syntax-semantic interface only. Still, the logical formula, and the form that the resolution restriction may not take, constrain the resolution: Some terms become unavailable as antecedents for a pronoun. We hence are able to make negative predictions – as the next sections illustrate.

4 Discourse and Bound-Variable Anaphora

We now illustrate logical formulas with pronoun terms and resolution restriction. We start with a series of simple examples of bound-variable anaphora, for various binding (quantification) scenarios.

For (5) below, (6) shows the logical formula with the pronoun term, and (7) the result of its Skolemization.

- (5) A man walked. A passer-by saw him.
- (6) $(\exists x. \text{Man}(x) \wedge \text{Walk}(x)) \wedge (\exists y. \text{PB}(y) \wedge \text{Saw}(f_1(y), y))$
- (7) $(\text{Man}(c_1) \wedge \text{Walk}(c_1)) \wedge (\text{PB}(c_2) \wedge \text{Saw}(f_1(c_2), c_2))$

The resolution of ‘him’ is represented by the (well-formed) restriction $f_1(c_2) = c_1$. More interesting is “A colleague left. He said he was ill.” whose Skolemized logical formula is

$$\text{Col}(c_c) \wedge \text{Left}(c_c) \wedge \text{S}(\text{I}(f_2), f_1)$$

Possible resolution restrictions are $f_1 = c_c$ and $f_2 = c_c$.

The example “A man walked. Every passer-by saw him.” illustrates that only existentials are subject to Skolemization; the universal quantifier is left as is, see [16]:

$$(\text{Man}(c_1) \wedge \text{Walk}(c_1)) \wedge (\forall y \in \text{PB}. \text{Saw}(f_1(y), y)) \wedge \forall y. f_1(y) = c_1$$

“A man walked. No passer-by saw him.” is analogous. (The existential formula under negation cannot be Skolemized, but it is not required in the example.)

On the other hand, “Every man walked. He whistled.”

$$(\forall x. \text{Man}(x) \rightarrow \text{Walk}(x)) \wedge \text{Whistled}(f_1)$$

has only the universal quantification, to which Skolemization does not apply. There is nothing therefore to identify f_1 with.

The logical formula for “No man walked. He whistled.” exhibits the existential quantification under negation, which cannot be Skolemized intuitionistically.

$$\begin{aligned} (8) \quad & (\neg \exists x. \text{Man}(x) \wedge \text{Walk}(x)) \wedge \text{Wh}(f_1) \\ & = (\forall x. \neg(\text{Man}(x) \wedge \text{Walk}(x))) \wedge \text{Wh}(f_1) \end{aligned}$$

Although it is intuitionistically valid to push the negation in, it results in a universally quantified formula, which is un-Skolemizable. Intuitionistically, negation is regarded as implication to $\perp$. The implication is dealt in §5 later, where we return to this example.

Negative universal quantification particularly clearly shows the merit of intuitionistically valid Skolemization. Classically, $\neg \forall x. \phi(x)$ is equivalent to $\exists x. \neg \phi(x)$, Skolemized as $\neg \phi(c)$. That would mean that “not every”, just like “some ...not”, permits cross-sentential anaphora, which goes against the usual judgement.³ However, the conversion from $\neg \forall x. \phi(x)$ to $\exists x. \neg \phi(x)$ is not intuitionistically valid: after all, although the proof of $\neg \forall x. \phi(x)$ implies there must be an interpretation of x for which $\phi(x)$ fails, the proof of does not reveal it. On the other hand, the proof of $\exists x. \neg \phi(x)$ does. Therefore, on intuitionistic Skolemization we distinguish “not every” from “some ...not”, as far as anaphora is concerned.

The example borrowed from [9] “Exactly one man loves Annie. He is rich.”, analyzed as

$$\begin{aligned} & \text{Man}(c_1) \wedge \text{Loves}(\text{annie}, c_1) \wedge \\ & \neg(\exists x. x \neq c_1 \wedge \text{Man}(x) \wedge \text{Loves}(\text{annie}, x)) \wedge \\ & \text{Rich}(f_1) \wedge f_1 = c_1 \end{aligned}$$

shows that our approach does not over-generate: it does *not* give a reading that there are possibly several men who love Annie, but only one of them is rich.

³ I thank Simon Charlow for this example.

Oleg: Here, we quantify under the believe. That isn't good. Its'e better to have $\text{Believe}(\text{lam}(s) \text{ Smart}(\text{Pr1}(x,s)),x)$

Intra-sentential bound-variable anaphora (9) is just as straightforward: (10), with the resolution restriction.

(9) Every male lawyer believes he is smart.

(10) $\forall x \in \text{MaleLawyer}. \text{Believe}(\text{Smart}(f_1(x)), x) \wedge \forall y. f_1(y) = y$

4.1 Quantificational Subordination

That (12) is not a felicitous continuation of (11) but (13) is is an example of quantificational subordination [9].

(11) Bryan bought a bottle of wine

at every store in the neighborhood.

(12) #It was a pinot noir.

(13) It was always a pinot noir.

The Skolem function for “a bottle of wine” must be an arity-1 function symbol, since it is in scope of “every store”. Whereas in (12) ‘it’ is represented by a constant symbol, in (13) it is also an arity-1 function symbol, being in the scope of the universal ‘always’. We may write the resolution restriction identifying the two symbols. Dropping “always” changes the arity of the pronoun symbol and makes the resolution restriction ill-formed.

The example shows there is room for refinement: more precise types. Consider (14) as the continuation for (11).

(14) ?It was a pinot noir from every country.

Here, the pronoun symbol for “It” is also of arity 1. Therefore, it is in principle possible to identify this symbol with the Skolem function for “a bottle of wine”. More precise types – not just arity 1, that is, $e \rightarrow e$ – would help. In (11), “a bottle of wine” is in scope of “every store” and is hence a function of stores. In (13), “It” is a function of events of buying wine, which refer to the the store the buying took place. Therefore, the “It” function can be converted to the function of stores. No such conversion is possible for (14), absent additional information (e.g., each store had wine from a different country).

5 Donkey Sentences and Local Skolemization

Donkey sentences are one of the main motivations for developing discourse representation theory and dynamic semantics. The discourse-anaphora donkey sentence “A farmer owns a donkey. He beats it.” [5, (1)] is straightforward, as expected. The famously problematic is the conditional donkey sentence “If a farmer owns a donkey he beats it.” – or a simplified version (15), see [9, (25)], whose logical formula may be written as (16):

(15) If Sarah owns a donkey, she beats it.

(16) $(\exists d. D(d)) \rightarrow B(f_1)$

where D is some predicate concerning Sarah and donkeys and B is about beating. Notably, the consequent is out of scope of the quantifier. The existential being on the left-hand-side of the implication, Skolemization does not apply either. We now show how the “existential disclosure” happens – naturally, without any extensions – and maintaining the scope of all quantifiers.

The solution is simple: observe that for any formulas ϕ and ψ

$$\phi \rightarrow \psi \equiv \phi \rightarrow (\phi \wedge \psi)$$

both classically and intuitionistically, which leads to

Proposition 1 (Skolemization of implication formulas). *If ϕ is Skolemizable (i.e., not under negation or at the left-hand side of an implication) then $\phi \rightarrow \psi$ is equi-satisfiable with $\phi \rightarrow S[\phi] \wedge \psi$*

Oleg: The equality seems to hold even we assume a modal semantics for conditionals, as reviewed in Higginbotham 2003, Conditionals and compositionality, Philosophical Perspectives: Language and Philosophical Linguistics, 17, 181-194.

As an example, (16) is equivalent to

$$(\exists d. D(d)) \rightarrow (\exists d. D(d)) \wedge B(f_1)$$

Indeed, both formulas express the meaning of the original sentence: either the set of donkeys Sarah owns is empty; or the set of Sarah’s donkeys and the set of things she beats are both non-empty. The

existential on the right-hand-side of implication (in the consequent) is Skolemizable. We thus obtain the equi-satisfiable

$$(\exists d. D(d)) \rightarrow D(c) \wedge B(f_1)$$

Just like in the original sentence, there is no guarantee that the sets of donkeys owned by Sarah and what she beats have anything in common. (After all, ‘it’ in (15) may well refer to something else, other than a donkey. Perhaps, owning a donkey predisposes a person to be violent.) However, the sentence does include the reading that what Sarah beats is the donkey. That is, the Skolem constants c and f_1 may be interpreted the same – or, the restriction $c = f_1$ may be imposed, giving the ‘donkey reading’ of the sentence. Although not explicit, identifying c and f_1 means that whatever interpretation of c satisfies $D(c)$, the same domain element – the same donkey – satisfies $B(f_1)$, i.e., gets beaten. We obtain thus the strong reading of the donkey sentence.

Although formally, the Skolem constant c is ‘global’ and hence one may refer to it outside the conditional, it is interpreted as the donkey owned by Sarah only when the condition is true. Otherwise, it is a random element of the domain. Thus, *semantically*, c is local, meaningful only within the conditional. As an illustration, we revisit (9) (repeated below as (17)) and write its meaning as (18): intuitionistically, negation is treated as implication to $\perp$.

(17) No man walked. He whistled.

(18) $((\exists x. \text{Man}(x) \wedge \text{Walk}(x)) \rightarrow \perp) \wedge \text{Whistled}(f_1)$

(19) $((\exists x. \text{Man}(x) \wedge \text{Walk}(x)) \rightarrow \text{Man}(c_1) \wedge \text{Walk}(c_1) \wedge \perp) \wedge \text{Whistled}(f_1)$

The formula can then be Skolemized as (19), letting as adjoin the resolution restriction $c_1 = f_1$. In some interpretation of (19), c_1 , and hence f_1 , may even refer to a man. However, as easy to see, no interpretation in which that man walks satisfies the formula.

6 Relative-clause Donkey Sentences and Binding out of DP

The other, just as challenging variety of donkey sentences is relative-clause sentences: (20). They present an additional puzzle, noted by

Charlow [3], who contrasted them with inverse linking: (21).

(20) $[_{DP}$ Every farmer who owns a donkey $_i]$ beats it $_i$.

(21) $[_{DP}$ Some kid from every city $_i]$ likes it $_i$.

Both sentences exhibit so-called ‘binding out of DP’ – specifically, the DP marked in the examples. The puzzle is that in (20), ‘it’ may refer to ‘donkey’ in case ‘a donkey’ scopes out of DP, *as well as* when the indefinite is confined within DP (that is, within the scope of ‘every farmer’). The meanings of course differ. On the other hand, in (21) the pronoun can be construed as bound by ‘every city’ only when ‘every city’ scopes above ‘some kid’. Charlow noted that Barker and Shan analysis of donkey anaphora [1] cannot reproduce the asymmetry: the very mechanism that lets Barker and Shan analyze donkey anaphora would apply to (21), and incorrectly predict inverse linking for the narrow scope of ‘every city’.

In our analysis, (20) has the following meaning on the wide-scope indefinite reading:

(22) $\text{Donkey}(c_1) \wedge \forall f \in \text{Farmer. Own}(c_1, f) \rightarrow \text{Beat}(f_1(f), f)$

The pronoun can then be resolved as $\forall f. f_1(f) = c_1$. The narrow-scope indefinite reading gives (23), which becomes (24) upon Skolemization:

(23) $\forall f \in \text{Farmer. } (\exists d. \text{Donkey}(d) \wedge \text{Own}(d, f)) \rightarrow \text{Beat}(f_1(f), f)$

(24) $\forall f \in \text{Farmer. } (\exists d. \text{Donkey}(d) \wedge \text{Own}(d, f)) \rightarrow$
 $\text{Donkey}(c_1(f)) \wedge \text{Own}(c_1(f), f) \wedge \text{Beat}(f_1(f), f)$

This case as well admits the restriction resolving the pronoun to the donkey – viz., $\forall f. f_1(f) = c_1(f)$, effectively identifying f_1 and c_1 – leading to the donkey reading.

In contrast, for (21), we obtain, for the surface and inverse scoping, resp.:

(25) $\text{Kid}(c_k) \wedge (\forall c \in \text{City. From}(c_k, c)) \wedge \text{Like}(f_1, c_k)$

(26) $\forall c \in \text{City. Kid}(c_k(c)) \wedge \text{From}(c_k(c), c) \wedge \text{Like}(f_1(c), c_k(c))$

Identifying ‘it’ with the city (that is, interpreting $f_1(c)$ as c) is indeed only possible upon the inverse scope. (25) has no existential to Skolemize.

The asymmetry noted by Charlow hence boils down to the fact that indefinite/existential QNP can be Skolemized but universal QNP cannot.

6.1 The Proportion Problem

The classical DRT analysis of donkey anaphora suffers from the so-called proportion problem [9, (28)]:

(27) Most women who own a donkey beat it.

According to DRT, the sentence is true if, of all pairs of women and their donkeys, the pairs in which the donkey gets beaten by its owner are in the majority. As [9] explains, if there are ten women of which nine own a donkey each and refrain from beating it, and one woman owns ten donkeys and beats them all, (27) would be predicted to hold, which seems clearly false.

On our analysis, (27) has the meaning

(28) $Most((\lambda w. \mathbf{Woman}(w) \wedge (\exists d. \mathbf{Donkey}(d) \wedge \mathbf{Own}(d, w))),$
 $(\lambda w. \mathbf{Beat}(f_1(w), w)))$

where $Most(S_1, S_2)$ is interpreted as true if $|S_1 \cap S_2| > 0.5 |S_1|$. (Here, $|S|$ is the cardinality of a set S .) Skolemization gives

(29) $Most((\lambda w. \mathbf{Woman}(w) \wedge (\exists d. \mathbf{Donkey}(d) \wedge \mathbf{Own}(d, w))),$
 $(\lambda w. \mathbf{Woman}(w) \wedge \mathbf{Donkey}(c_1(w)) \wedge \mathbf{Own}(c_1(w), w)$
 $\wedge \mathbf{Beat}(f_1(w), w)))$

to which the resolution constraint $\forall w. f_1(w) = c_1(w)$ may be adjoined. That is, the sentence is true just in case the subset of donkey-owning-and-beating *women* is the majority in the set of donkey-owning women – which is the expected meaning.

7 Discussion and Related Work

We have concentrated in the paper on the way to represent not yet resolved – and later, resolved – pronouns in logical formulas that express the meaning of a sentence, and how it may constrain the

pronoun resolution. Representing not-yet resolved pronouns and explicating the semantic constraints on their resolution has also been one of the main goals of DRT [8]. DRT (and the related [11]) approach considers the resolution an integral part of the semantics: coindexing on the fly. We, on the other hand, regard the resolution as a separate phase, after the complete semantic analysis: so-called external coindexing. Context-update-driven reference change is likewise out of scope for us. As an important technical difference, we represent not yet resolved pronouns not by (anaphoric) discourse referents (DRT), reference markers (in early DRT) or variable indices or registers (in [11]) – but Skolem *functions*. Therefore, we are able to account for the dependence of these not-yet-resolved pronouns on the variables in scope.

We leave coindexing to the separate pragmatic step because it is inherently context and world-knowledge dependent. The following complete paragraph is a good illustration. There are many occurrences of ‘he’ and ‘him’, whose antecedents are not always the same people and occur sometimes later in the discourse; ‘this’ and ‘it’ are similarly ambiguous and require world knowledge to resolve.

“Now just one thing more remained, the box that held the daylight, and he cried for that. His eyes turned around and showed different colors, and the people began thinking that he must be something other than an ordinary baby. But it always happens that a grandfather loves his grandchild just as he does his own daughter, so the grandfather felt very sad when he gave this to him. When the child had this in his hands, he uttered the raven cry, ”Ga,” and flew out with it through the smoke hole.”

“Raven”, Tlingit Indians of Southeastern Alaska⁴

In a short paper [11] Muskens essentially argues (albeit not saying so directly) that coindexing-first theories may be regarded as non-deterministic specifications, in the spirit of Rabin and Scott [14]. The non-determinism is eventually resolved via some algorithmic procedure: for example, guess-and-backtrack or constraint-propagation, as in Prolog or constraint logic programming.

⁴ <http://www.ankn.uaf.edu/IKS/subsistence/Tlingit/myths.html>

There is also a precedent of using choice functions – another name for Skolem functions – for scope and anaphora of indefinites, among others [6, 17]. As formulated, that approach requires second-order logic.

Predicate Logic with Anaphora (PLA) was proposed by Dekker [5] as a compositional and rigorous formulation of discourse representation theory with the fully orthodox notions of scope and binding (unlike its predecessor Dynamic Predicate Logic (DPL) [7]) and as a proper, *dynamic*, extension of ordinary predicate logic. Its main idea is treating (nominal) pronouns as a separate category of terms, distinct from both constants and variables. In stark contrast to DPL, pronouns are not subject to binding. However, PLA is predicated on the pronouns being already resolved. Like DPL, PLA is a dynamic logic theory through and through.

Dekker paper [5] has also identified the principal stumbling block for any logical theory of discourse anaphora: if an antecedent for a pronoun is an indefinite in an earlier sentence in the discourse and the sentence is represented by a closed existential formula, there is no way to refer to that antecedent. We would need an “existential disclosure” so to speak [4]. Dynamic semantics is one way of achieving such disclosure; Σ -types in Dependent Type Semantics is another. (Dynamic semantics and dependent types may, of course, have other uses.)

PLA and its exposition of the stumbling block has been the inspiration for this work, stimulating the search for the minimal workaround. We also wanted to avoid the baking-in of the pronoun resolution, which is responsible for many PLA problems: inability to write the denotation of a sentence in isolation, the failure of weakening, the entailment relation generally being not transitive or even reflexive, and the need for “some additional bookkeeping”. Like in PLA, but clearly unlike DPL, we represent pronouns by constants – not variables. Unlike PLA, our constants are ordinary, rather than a special category with a special, dynamic, treatment within a model.

Like us, Butler (in the long line of research, see [2] for the latest) treats antecedent and pronouns as entities, to be identified in the resolution procedure. Unlike us, the resolution procedure is one of the main topic of Butler’s research and described in great detail. Butler also employs Skolemization as an intermediary step.

Dekker [5] has noted that “in almost all compositional approaches to anaphora, pronouns are associated with (syntactically free, but semantically somehow bound) variables.” His proposal, PLA, is similar despite the claims to the contrary: PLA associates pronouns with dynamically bound [10], rather than ordinary lexically-bound, logic variables. Our approach, in contrast, treats pronouns as terms headed by a fresh function symbol – quite similar to a Skolem term.

Unlike Barker and Shan [1], we do not need nor want to treat the implication $\phi \rightarrow \psi$ as $\neg\phi \vee \psi$, or $(\exists x. \phi) \rightarrow \psi$ as $\forall x. \phi \rightarrow \psi$, which are not valid intuitionistically. We hence maintain the scope of quantifiers and island effects. Our approach applies even when “if” is not taken to be the classical material implication.

8 Conclusions

We have described how the ordinary orthodox predicate logic from 1920s by itself, without any extensions, with the standard quantification and binding, represents a variety of pronominal discourse, bound-variable and donkey anaphora. Specifically, we have described the way to represent not yet resolved pronouns in logic formulas, and then record the result of their resolution – also as a logic formula that effectively identifies the pronoun with its antecedent. Although the mechanics of finding the antecedent is delegated to the next, pragmatic phase, the formula helps the resolution by constraining the choices of antecedents. We hence obtain predictions, of unavailability of particular antecedents and readings.

The approach is inspired by, and extensively uses the standard (intuitionistic) Skolemization. We stress intuitionistic validity and hence avoid classical steps (e.g., conversions of implications) that leave no witnesses. Intuitionistic Skolemization is also important to avoid overgeneration.

For future work, we would like to analyze further anaphoric phenomena: paycheck pronouns, epistemic modalities, split antecedents, telescoping and parascopeing.

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